1 Magnetic field control

The magnetic field control system consists of the magnetic shield, the set of shim coils, and the internal and external magnetometers. Together, these components allow for interaction region operation with low levels of ambient magnetic background present. This section will discuss the design requirements of the system, present the physical details of its implementation, as well as the performance achieved with the present shielding system.

While the design, building, and characterization of the shield can be done in a straightforward manner by emulating past work, the underlying science is frequently not obvious and is often left out of the literature on the topic. This is understandable, since understanding how the shield works is rarely of interest to the experimentalist desirous of getting an atomic or molecular beam operational. In what follows, I will also have to leave several lines of exploration unfinished as they are not immediately required in order to proceed with the CENTREX experiment. However, it may be a starting point for future work requiring higher performance.

1.1 Design requirements

The allowed transverse residual field perpendicular to the interaction-region electric field $\mathbf{E}$ is limited by the motional magnetic field experienced by the molecules. As they fly through with a velocity $\mathbf{v}$, they experience the motional field $\mathbf{B}_{\mathrm{mot}} = \mathbf{v} \times \mathbf{E}/c^2$ that changes direction when the electric field is reversed. Any residual field in the same direction will add to $\mathbf{B}_{\mathrm{mot}}$, such that the magnitude of the total field will change with electric field reversals. Due to the Zeeman splitting between the spin up and down states, the change in magnetic field magnitude introduces a frequency shift.

With our experimental configuration, $|\mathbf{E}| \approx 30$ kV/cm and $|\mathbf{v}| \approx 180$ m/s, the magnitude of the motional field is $|\mathbf{B}_{\mathrm{mot}}| \approx 60$ μG. On the other hand, the Zeeman shift is approximately 50 nHz/μG, so the maximum change in magnitude of the total magnetic field corresponds to approx. 3 μHz, which is 60$\times$ above the anticipated statistical sensitivity of the experiment [**?** , sect. 3.1]. However, due to M modulation, the shift is further suppresed by an order of magnitude, such that the residual fields need only be a factor of 6$\times$ lower

than the motional field; that is, the average (why average? why not rms, or something else entirely?) residual fields of up to 10 μG are allowed.

The requirement on the other two components of the static residual field is much less strict, since it does not depend on the direction of the electric field and will get cancelled by averaging over the E reversals. Still, how much can we tolerate?

The condition for dynamic shielding is to suppress the high-frequency (i.e., higher than the timescale of E-field reversals— though what's the exact bandwidth over which it matters?) lab magnetic noise enough so it does not Zeeman shift the measurements considerably more than shot noise limited sensitivity. Along the electric field, the slope of the resonance frequency with respect to the external B field is $|\mathrm{d}f/\mathrm{d}B| \approx 4.0$ mHz/μG, while the shot-noise limited sensitivity is 0.7 mHz. Thus an rms (rms or average or something else?) internal field noise of 0.2 μG can be tolerated; if the lab noise is estimated as 1 mG, then the required transverse shielding factor is of the order of 5,000.

The noise along the other transverse direction, and the axial noise, is still less strictly constrained. The Zeeman shifts due to fields perpendicular to the electric field are quadratic and of the order of 0.7 mHz/mG^2. This means than dynamic shielding is not even required along these directions.

1.2 Theory and related work

Paramagnetic shielding

Degaussing Degaussing is a method of reducing the residual field inside a magnetic shield based on the following observation: the field inside a hollow cylinder made of a ferromagnetic material, when placed in a region of a given external field, can be greatly reduced by driving a sinusoidal magnetic field around it with a gradually decreasing envelope.

A common explanation for the decrease in the internal magnetic field is as follows. The sinusoidal magnetic field drives the ferromagnetic material into saturation. Regardless of the inital magnetization of the material, it will move around its hysteresis curve (see Fig. 1a) and lose all memory of the initial condition. The external magnetizing current is applied symmetrically around zero (necessitating some form of DC-blocking, such as a

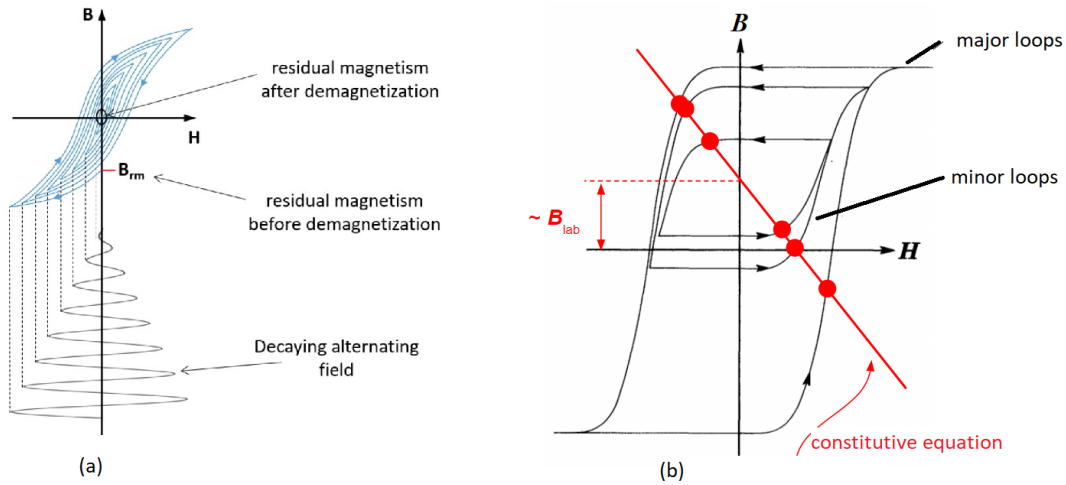

Figure 1: ***a.*** Degaussing: naive picture. ***b.*** Major vs minor loops.

transformer or capacitive coupling), so when the amplitude is gradually decreased to zero, no net magnetization will remain (e.g., [**?** , sect. 3.4.2]).

Unfortunately, this explanation only applies for when a single grain of the material is degaussed in a region with no net external field—hardly applicable to magnetic shielding. In practice, the shield would be situated in a region with significant external field, such that the total externally applied field is *not* symmetric upon current reversal. Similarly, over the course of the decrease of the current, the magnetization would ideally settle on a value that as much as possible *opposes* the stationary lab field. Rather than equating degaussing with a demagnetization process, it would be more appropriate to consider it a particular case of inducing a non-zero magnetization in the material; in fact, a perfectly demagnetized shield would be perfectly useless.

It can be enlightening to think about the shield in terms of an equivalent current distribution. Given some distribution of magnetization $\mathbf{M}(\mathbf{x})$ at position $\mathbf{x}$ inside a volume V bounded by a surface S, the corresponding magnetic vector potential is [**?** , eqn 5.103]

$$\mathbf{A}(\mathbf{x}) = \frac{\mu_0}{4\pi}\int_V \frac{\nabla' \times \mathbf{M}(\mathbf{x}')}{|\mathbf{x}-\mathbf{x}'|}\mathrm{d}^3x' + \frac{\mu_0}{4\pi}\oint_S \frac{\mathbf{M}(\mathbf{x}')\times\mathbf{n}'}{|\mathbf{x}-\mathbf{x}'|}da',$$

where $\mathbf{n}$ is the normal to the surface S at point $\mathbf{x}$. Considering a layer of magnetic shielding as a very thin cylindrical shell, the first term has a negligible contribution assuming the magnetization does not possess a pathologically large curl. The second term is the 'surface

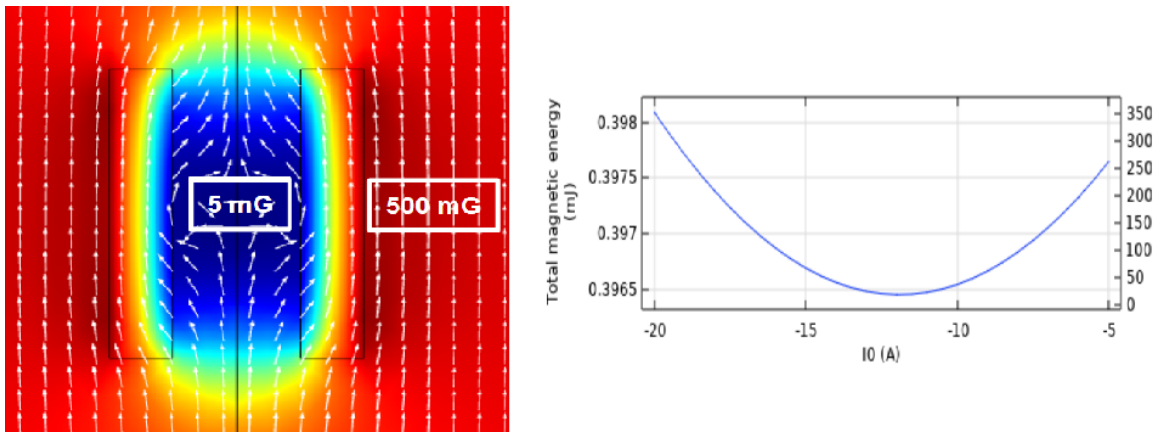

Figure 2: Toy model for a degaussed shield: ***a.*** cylindrical shell with a solenoidal current distribution. ***b.*** Total magnetic energy as a function of the solenoidal current.

current' term if we identify $\mathbf{K}(\mathbf{x}) = \mathbf{M} \times \mathbf{n}$ as the surface current.

As a simple example, for a cylinder uniformly magnetized along its length, the surface current is perpendicular to the surface normal, as well as the axis of the cylinder; i.e., it is equivalent to a solenoid. Given that a properly degaussed cylindrical shield opposes the lab field, it must therefore be magnetized in a direction opposite to the external field, or equivalently, possess a non-zero current distribution that in some way opposes and thus attenuates the lab field. (A demagnetized shield, having a zero surface current opposing the lab field, would obviously not work as a shield.)

Any magnetic shield can be modelled as a suitably magnetized shell surrounding the region of interest for an experiment. Of course, we cannot in general expect uniform magnetization (i.e., perfectly solenoidal surface currents) to result from a degaussing procedure applied on a cylindrical shell. While the form of the current distribution obtaining after degaussing is left as an exercise for future research, I can note that it is likely determined by a combination of geometric factors (e.g., constraining the spatial current distribution to a cylindrical shell), and the details of the motion of hysteresis loops of a given material. In the following two sections, I will draw attention to some of the features of these factors.

Degaussing toy model To find what happens to the magnetization of the shield during degaussing, we can begin by recalling that any system will likely arrive at the state of lowest energy it can reach, so long as it has enough activation energy to leave its current

local minumum. Given that the magnetic flux density B corresponds to a magnetic energy density $u = B^2/2\mu_0$, it is reasonable to expect that the shield is striving to minimize the magnetic field inside it, though subject to two constraints.

On the one hand, the magnetic domains are not entirely free to reorient, as they may get pinned at material imperfections; degaussing helps with this issue by moving the domain walls to reach a more optimal configuration with respect to minimizing the total energy. Thus we may expect that the more of these imperfections, the less effective the degaussing; hence the importance of annealing magnetic shields and handling them carefully. On the other hand, given that magnetic field lines are closed loops, the field inside the shield that opposes an external field eventually loops around in such a way as to increase the total field outside. Thus even perfect degaussing is unlikely to result in a zero magnetic field inside the shield.

To illustrate, consider the Comsol simulation of a cylindrical shell with solenoidal current placed in an area of a non-zero external magnetic field (Fig. 2). By setting the current to a value that minimizes the total magnetic energy, we obtain a situation that resembles the excellent shielding characteristic of degaussed shields, complete with the small non-zero field inside the shield. In this case, the current distribution was constrained to be a simple solenoid. One can nest concentric solenoids to model a multi-layer shield and we find that in the minimum energy configuration, the inner layer has a zero current distribution, i.e., it does not participate in the shielding. This is consistent with the experimental observation that once a shield layer has attenuated most of the lab field, adding further layers does not improve the residual field.

Further work may lift the solenoidal limitations of this toy model and consider the case of a general surface current distribution that minimizes energy. Together with some model of hysteresis behaviour (next section), one could come up with an explanation for the performance limits of degaussed magnetic shields.

Hysteresis during degaussing Referring to Fig. 1b, we can qualitatively identify two distinct groups of hysteresis loops. Major loops are those where the material is driven to saturation on both sides of the axes. These are thought to be unique for a given material

and repeatable, since the material starts and ends each time in the same configuration (having all domains oriented as much as possible in the same direction). On the other hand, minor loops are those where the amplitude of the external driving is too small to result in magnetic saturation. [?] Even under constant external excitation, these loops can drift around. Different loops can intersect or share sides, since the same $(\mathbf{H}, \mathbf{B})$ point in the diagram can refer to different microscopic configurations. [?] The drift of minor hysteresis loops can reflect the growth of microscopic patterns of magnetic domains [?].

Using this terminology, we can provide a better description of what happens to the ferromagnetic material during degaussing. First, the large external field drives the material around the predictable major loops. As the amplitude decreases, the material is forced to spiral around a non-trivial set of minor loops with the intention of arriving at low B values. By the time the external field stops changing, the material has come to a rest at a point on the last minor loop, at its intersection with the constitutive relation between $\mathbf{B}$ and $\mathbf{H}$ (solid line in Fig. 1b).

The constitutive relation can easily be found from classical magnetism [? , sect. 5.11]. Consider a ferromagnetic object magnetized with a magnetization M, which results in a field $\mathbf{B}_{\text{obj}}$. The auxiliary field is

$$\mathbf{H}_{\text{obj}}(\mathbf{M}) = \mathbf{B}_{\text{obj}}/\mu_0 - \mathbf{M}.$$

If we apply a constant external field $\mathbf{B}_{\text{ext}} = \mu_0 \mathbf{H}_{\text{ext}}$ throughout all space, the total field is

$$\begin{aligned} \mathbf{B}_{\text{tot}} &= \mathbf{B}_{\text{ext}} + \mathbf{B}_{\text{obj}}(\mathbf{M}) \\ \mathbf{H}_{\text{tot}} &= \mathbf{H}_{\text{ext}}/\mu_0 + \mathbf{H}_{\text{obj}}(\mathbf{M}) \end{aligned} \tag{1}$$

The system in eqn 1 gives one relation between $\mathbf{B}_{\text{tot}}$ and $\mathbf{H}_{\text{tot}}$. As mentioned, the other degree of freedom is fixed by the final hysteresis loop, which is significantly more challenging to model [?]. Recent work used neural networks as universal approximators to predict hysteresis behavior [?]; in particular, a recurrent network has been used to model time-dependent hysteresis effects [?]. Some of these models have been applied to predict the value of the magnetic field inside a three-layer mu-metal shield using a magnetometer

placed outside the shield for a space-based atomic clock [**?**], to model shield degaussing [**?**], or to investigate the theoretical limit of degaussing in magnetic shields [**?**].

1.3 Physical structure

The magnetic shield consists of four concentric cylindrical layers with open ends (See Figs. 3 and 4). Three of these are built by wrapping metglas foil around PVC pipes, while the fourth, innermost layer has been welded out of two sheets of mu-metal rolled into a tubular form, and annealed. The mu-metal tube, being 1 mm thick, is relatively fragile; in order to limit its motion, it is sandwiched between the hard plastic PVC tubes of the inner metglas layer and that of the shim coils. The inner two metglas layers are spaced apart using small-diameter PVC tubing; the same method is employed to suspend the glass vacuum chamber of the interaction region centered inside the shim coils tube. The outermost tube is mounted on an 80/20 frame using a set of seven aluminum semi-circular plates.

Layers, spacers, mounting The three metglas layers are all built in the same way. First, the plastic tube (provided by the Harrison Machine & Plastic Corp., Garrettsville, OH) is cut to size with a handsaw if necessary (see Table I for layer lengths and material details) and roughly cleaned with water and paper towels on the outside and, as far as possible, also on the inside to remove the dirt it comes with, probably due to outdoors storage. Then, I cut 8×12 pieces of metglas (7×12 for the 16" layer since it is 15" shorter than the other two) to the length required to wrap around the tube with approx. 1" of overlap. The pieces are secured to the PVC tube and to each other with a few short pieces of Kapton tape (McMaster p.n. 7648A63) so that they do not fall down during assembly.

The prepared and cleaned tubes are put on a stand with wheels under them so they can be rotated, facilitating the circumferential wrapping of the metglas foil. The first piece of tape is wrapped about an inch from the end of the tube; leaving this much bare PVC helps with handling the approx. 100–200 lb tubes. The second piece is wrapped so as to have about an inch of overlap with the first, and the remaining six pieces are attached analogously. Once the first layer is done, the wrapping resumes at the position of the first metglas piece. This procedure is repeated for a total of 12 layers total. (Due to axial and

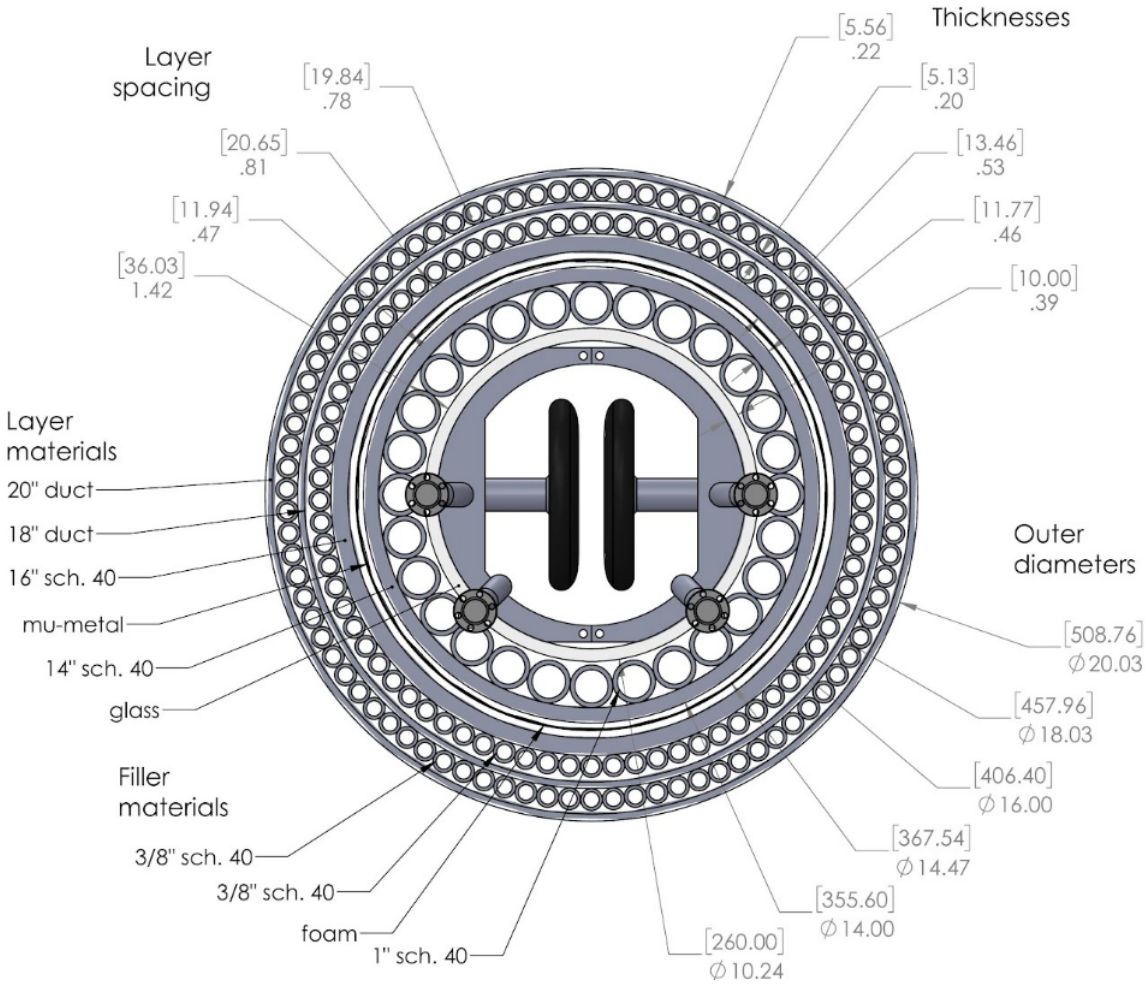

Figure 3: Cross-sectional view of the magnetic shield.

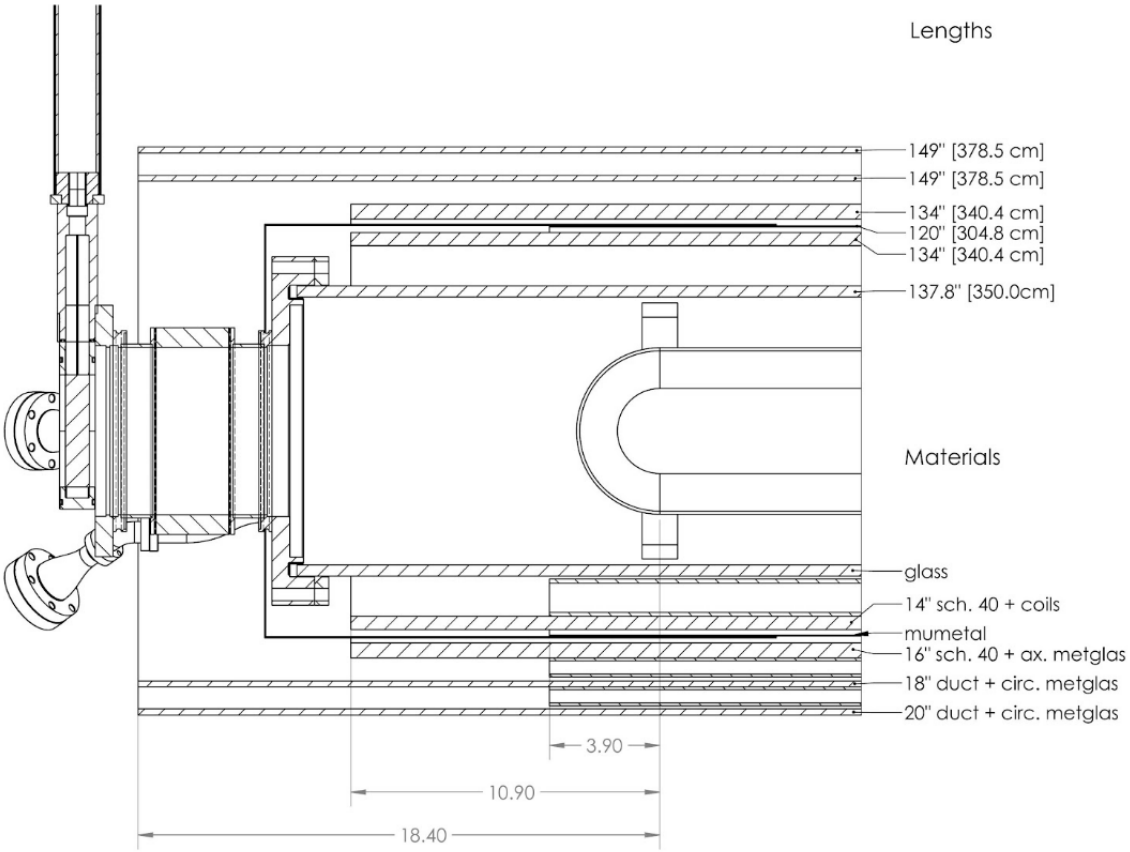

Figure 4: Longitudinal cross-section of the magnetic shield, showing the unequal lengths of the layers.

Table I: The four layers of the magnetic shield.

Layer	Length [in]	Weight [lbs]
Main layers:		
20” duct	149	100
18” duct	149	80
16” sch. 40	134	180
14” sch. 40	134	140
Spacers:		
3/8” sch. 40	120*	190
1” sch. 40	120*	90

circumferential overlaps, the true thickness of the shield is not perfectly homogeneous, but in practice we find that this is unimportant as long as the overall shielding is sufficiently high.)

To secure the metglas pieces and prevent them from coming loose in case the Kapton tape fails, the entire tube is wrapped in plastic foil. Following that, there are 12 turns of degaussing wire wrapped in a toroidal fashion around the tube. The wire is 22 or 24 gauge lacquered copper wire, sufficiently thick for approx. 2 amps of degaussing current. The 12 turns are uniformly distributed around the circumference of the tube; we believe this may be important for the performance of the degaussing. On the outermost (20”) tube only, we also have a solenoidal coil with approx 1 turn per in spacing for rough cancellation of axial fields. (Given the metglas wrapping geometry, this solenoidal coil would not be effective for degaussing.) Finally, each tube is again wrapped in plastic wrap to secure the degaussing and (for 20” tube only) the solenoidal coil.

The mu-metal layer does not, of course, require wrapping with metglas, only the degaussing coil and the outer plastic foil wrap. However, care must be taken to reinforce the insulation of the degaussing wire at the ends of the cylinder; the sharp edges can easily strip the magnet wire coating and short the turns of the wire to the conductive cylinder, which does not happen with the PVC tubes. I used heat shrink tubing to do this, but it required cutting the wire at each turn and at both ends, adding insulation, soldering the wires back together and heating the insulation to make it shrink and adhere. It would have been better to use a wire with more durable insulation instead.

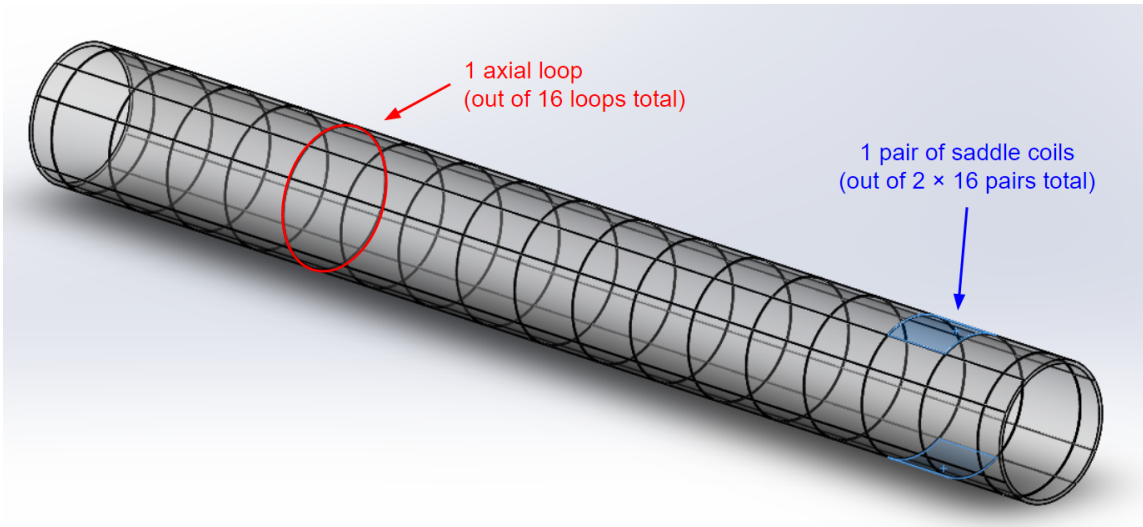

Figure 5: Shim coil geometry.

Shim coils The set of shim coils consists of 16 segments of three dimensions each (see Fig. 5). The two transverse dimensions (B_x and B_y) are implemented as pairs of one-loop saddle coils (8" by 60°each), while the axial field is controlled by simple circular current loops (spaced 8" apart from each other). This gives a total of $5 \times 16 = 80$ degrees of freedom. The 22g magnet wires are fixed in place by being glued into a series of grooves on the surface of the 14" PVC tube using neutral-cure RTV silicone sealant (Dow Dowsil 737) and wrapped in plastic foil for added protection. During the curing of the silicone, the wires are held in place by Kapton tape, which is removed after the silicone hardens.

The size of the tube makes it impractical to use standard milling machines. While the circular grooves can easily be cut using a hand saw, the axial ones were done on a custom-made setup (Fig. 6) involving a hand drill clamped on a 80/20 frame, which in turn is mounted using linear bearings to a larger frame. The setup is not as rigid as a true milling machine would be, which is why it is important to use a constant force while moving the carriage forward to avoid cutting wavy grooves. I used a system of ropes, pulleys, and weights to achieve satisfactory results.

With 80 different pair of wires (twisted into twisted pairs using a hand drill), cable management is essential so as not to confuse the cables with each other. For each coil, a linear groove is cut from one end of the tube to the location of the coil; thus all coils can be connected to at one end of the shield. There, the sets of wire pairs belonging to each

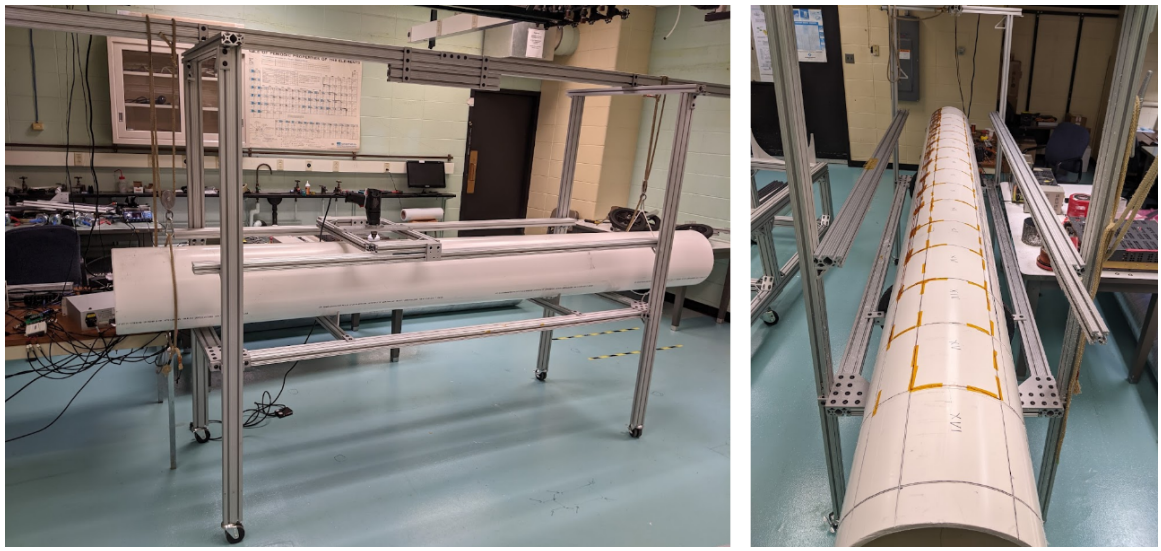

Figure 6: ***a.*** Milling machine for cutting longitudinal grooves. ***b.*** Wires taped inside grooves while silicone hardens.

dimension are bundled together into five bundles of 16 pairs using zip ties; each wire pair is temporarily labelled by a numbered piece of Kapton tape at the end. For ease of connecting and disconnecting, these wires are attached to adapter boards that are permanently attached to the 80/20 support frame of the shield. The adaptor boards bundle four pairs of wires into one RJ45 jack to enable using standard patch cables for ease of connection.

Support structure The considerable total weight of the magnetic shield (780 lbs of plastic, plus metglas, mu-metal, and other materials) requires a sufficiently strong support structure. In our case, it is assembled from 80/20 rails (profile number 3030) and placed on eight caster wheels McM p.n. 9813T71) so the setup can in principle be moved by one person. The support structure consists of three four-foot segments mounted together to form a rectangular box. Rails going across the top of the box hold aluminum plates (1/4" thick) with semi-circular cutouts. To keep the plates from damaging the delicate outer metglas layer, the sharp corners of the cutouts are protected with hard nylon tape.

1.4 Operation and measurements

The characterization and use of the shield is an iterative process requiring a repeated sequence of three operations: magnetic field measurement, degaussing, and shimming.

I distinguish two types of measurement of the magnetic field of interest to us. Residual field is the field inside the shield assuming no changes in the external magnetic conditions. In our four-layer shield, this is the dominant magnetic field inside the interaction region and it is driven primarily by the interaction between the magnetization of the ferromagnetic shield layers with the external magnetic field. Despite our best attempts at degaussing, these residual fields are typically of the order of 50–200 μG in the center of the shield.

Residual field measurement During shield characterization, the primary measurement is of the field in the central region of the shield. During the use of the shield in a beam-line experiment, this region is difficult to access; it is also the region of greatest interest since all molecules included in the experiment pass through it. During characterization I attached a 4" $\times$ 4" aluminium rectangular tube by suspending it from them shield mounting structure at each end. I used a system of pulleys to pull the sensor carriage using a stepper motor and a rotary encoder to measure the position of the sensor along the length of tube. The sensor itself (Bartington Mag03MS-100) is a 3-channel fluxgate magnetometer with a measuring range of 1 G and a typical electronic offset of several hundred microgauss. (The specified noise level of 0.1 μG/$\sqrt{\text{Hz}}$ at 1 Hz, integrated over the 3 kHz bandwidth of the sensor, is below the mechanical stability of the sensor translation setup with a typical repeatability of approx 2 μG.)

The electronic offset of the sensor and its associated electronics is significant. It is specified as 50 μG, with a temperature coefficient of 1 μG/ °C, although in practice offsets of several hundred microgauss are typical. The offset calibration is done by taking a measurement and immediately repeating it with the sensor rotated upside down. This is accomplished by mounting the sensor in a short rectangular aluminum tube with enough plastic tape on the outside so that the sensor carriage fits tightly inside the long square tube going through the center of the shield. After a measurement is done, the sensor is pulled out of the long tube and reinserted rotated upside down. The accuracy of the offset calibration then depends on how well-centered the sensor is in the carriage, how much the field inside the shield changes during the time it takes to perform the calibration, how much the electronic offset itself changes in the same time, and how perfectly square the cross-sections

of the long tube and the carriage are.

Rather than attempt to estimate the magnitudes of these effects individually, I estimate the precision/accuracy of the calibration from the calibration curve itself; see Fig. 7 for a typical example. It shows the two transverse components of the magnetic field (B_x and B_y) as measured by the sensor in its normal orientation, as well as with the sensor inverted. The data is presented in the symmetrized form: (normal + inverted)/2, which corresponds to the sensor offset, and (normal − inverted)/2, which corresponds to the physical magnetic field plus any calibration imperfections. Given the time scale of the changes in the electronic offset is primarily driven by temperature changes, which are negligible in the time it takes to do the calibration (around two minutes), we expect the curves showing the offset to be perfectly flat within the noise of the measurement, but that is typically not the case. The RMS deviations from the average value I identify as the error in the offset calibration, consisting of all the aforementioned factors.

While one may take the (normal − inverted)/2 superposition as the best measurement of the physical fields, doing so means that the error is different at each location along the shield. Since we expect the offset to be essentially constant during the measurement, I believe it a better practice to estimate the offset from the average value of the superposition, and then simply subtract this constant value from all subsequent measurements. This way, the overall offset is subject to an uncertainty, but the shape of the field pattern is known much more precisely.

The typical repeatability of a measurement can be within ± 1 μG, as shown in the example measurement in Fig. 9. In addition to the random errors introduced by the mechanical instability of the measurement setup, we also have to pay attention to the systematic errors induced by the signal chain from the sensor onwards. The sensor and its preamplifier have a limited bandwidth; 3 kHz for the sensor, and a settable low-pass filter for the preamp. This must not be set too low in relation to the speed of the sensor motion. In practice, I use the lowest reasonable LPF cutoff frequency, verifying the choice by repeating the measurement with the next higher available frequency. If the two measurements are sufficiently close to each other, I use the lower of the two frequencies. I use a similar process to decide upon the digital LPF included with the ADC.

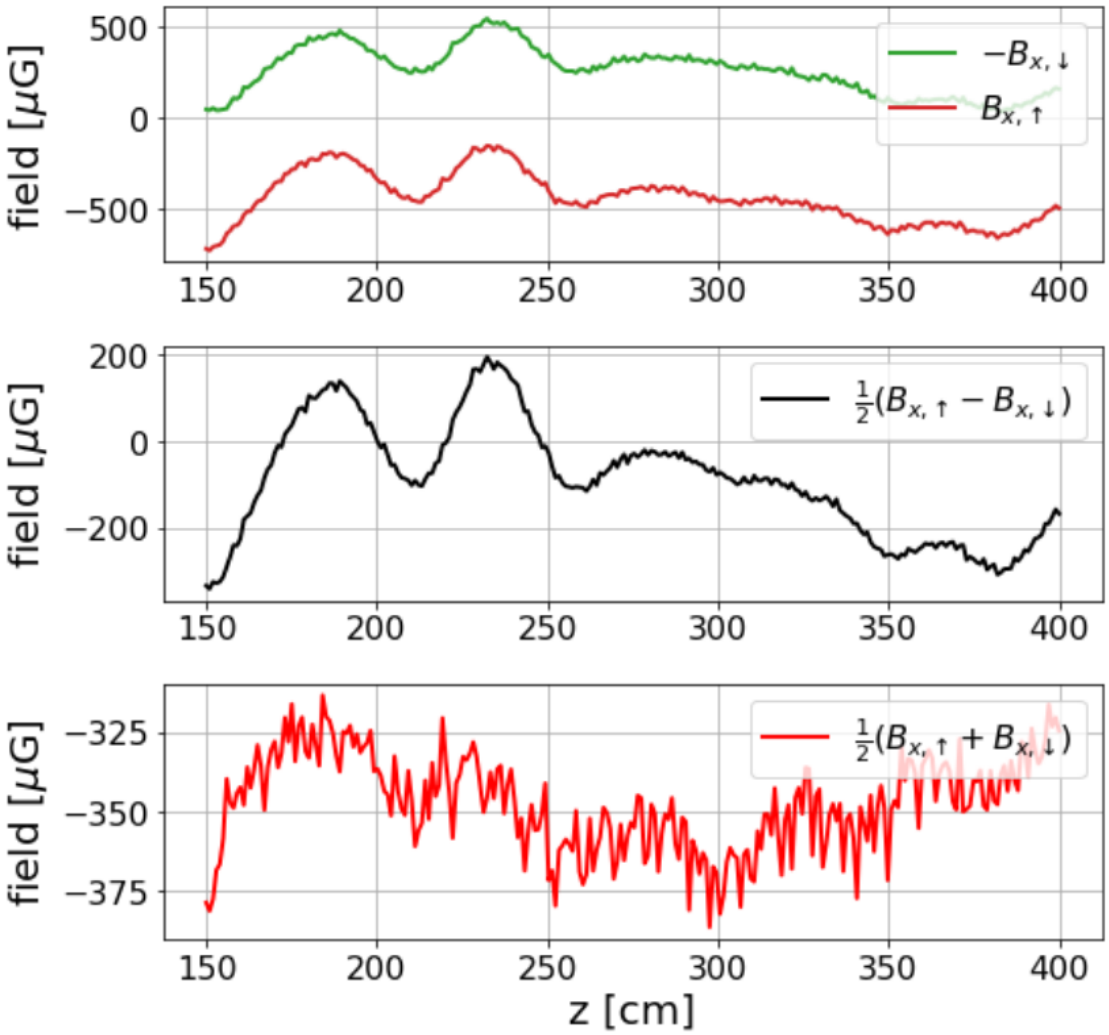

Figure 7: Electronic offset of the magnetometer as a function of position along the length of the center of the magnetic shield. Only the two transverse dimensions are shown.

Dynamic field measurement The attenuation of the external fields is not perfect. If an infinitely large Helmholtz pair were to drive a perfectly uniform and sinusoidally varying magnetic field in the region of space occupied by the magnetic shield, then a smaller sinusoidal variation of the field inside the shield could be detected that is of the same frequency as the external field. In practice, there are practical limits to the size of exteral coils. In one version of the measurement, I have attached a room-sized loop of wire to the walls of the lab and drive them with a function generator while passing the magnetometer reading through a lock-in amplifier. Repeating the measurement in the absence of any shielding gives the magitude of the external field variation. Dividing this by the magnitude measured with the shield in place gives the shielding factor.

The disadvantage of this method is that the magnetic shield has to be removed to measure the external magnetic field, since the shape of the external coils is difficult to measure precisely. (Future work: build room-sized coils of known geometry to measure SF better.) To find the SF of the final shield, I have used smaller rectangular coils of well-known dimensions centered around the magnetic shield; the field they apply, while not uniform, can be calculated using a closed-form expression. This allows me to define an effective shielding factor as the ratio of the theoretically predicted field of the coils divided by the field measured inside the shield.

I used the SR530 analog lock-in amplifiers to measure the internal field magnitude. In order to extend the measurement below the 0.5 Hz limit of this instrument, I have also tried to measure the fields inside with the ADC used in residual-field measurement, but without any further preamplification and filtering. However, I have not yet succeeded in reconciling these measurements with those done with the SR530. In particular, the software lock-in reported non-zero internal field at zero external excitation. In the future, I may verify why that is the case, and spend more time calibrating the software lock-in as compared to commercial instruments.

To calibrate the externally applied field, I measured the current flowing through the rectangular coils by measuring the voltage drop over as series 1 Ω resistor with an oscilloscope. At low amplitudes, this measurement is susceptible to noise picked up by the wiring because there is no filtering between the resistor and the oscilloscope. This could

be improved by using a lock-in amplifier to calibrate the current versus function generator amplitude and frequency settings.

Off-axis magnetometry When the magnetic shield is in use, the central axis will not be accessible to magnetic sensors. To control the magnetic field, we plan on using a set of four magnetometers distributed around the vacuum chamber. These are currently in an aluminum slide, but since they are fluxgate sensors, the dynamic field generated by the sensor may induce currents in the aluminum and cause spurious measurements. I may have to move the sensors into plastic slides instead.

Averaging in frequency domain. Consider a physical signal $V(t)$ that we measure for a total time T from $t = -T/2$ to $t = T/2$. The measured signal thus is

$$V_{\text{meas}}(t) = V(t) \cdot \Pi(t/T),$$

where $\Pi(t)$ is the rectangular function

$$\Pi(t) = \begin{cases} 0, & |t| > 1/2 \\ 1/2, & |t| = 1/2 \\ 1, & |t| < 1/2. \end{cases}$$

The spectrum of the measured signal is given by the Fourier transform:

$$\begin{aligned} \widetilde{V}_{\text{meas}}(\omega) &= \mathcal{F}[V_{\text{meas}}(t)] \\ &= \mathcal{F}[V(t) \cdot \Pi(t/T)] \\ &= \mathcal{F}[V(t)] * \mathcal{F}[\Pi(t/T)] \\ &= \widetilde{V}(\omega) * \text{sinc}(\omega T/2). \end{aligned}$$

Thus the frequency-domain effect of measuring a signal for a time T is a convolution of the signal's spectrum $\widetilde{V}(\omega)$ with the sinc function.

Now consider dividing the measurement into N equal segments $V_k(t)$ of length $T_0 = T/N$, where $k = 1, \ldots, N$. These segments correspond to time-shifting the original signal $V(t)$, then truncating it to a length T_0:

$$\begin{aligned} V_k(t) &= \Pi(t/T_0) \cdot V(t - T/2 + T_0(k - 1/2)), \quad k = 1, \ldots, N \\ &= \Pi(t/T_0) \cdot V(t - \Delta t_k), \quad \text{where } \Delta t_k = T/2 - T_0(k - 1/2). \end{aligned}$$

If we average these N segments together, we get the averaged signal

$$V_{\text{avg}}(t) = \frac{1}{N} \sum_{k=1}^{N} V_k(t) = \frac{\Pi(t/T_0)}{N} \sum_{k=1}^{N} V(t - \Delta t_k).$$

The spectrum of the averaged signal is

$$\begin{aligned} \widetilde{V}_{\text{avg}}(\omega) &= \mathcal{F}[V_{\text{avg}}(t)] \\ &= \mathcal{F}\left[\frac{\Pi(t/T_0)}{N} \sum_{k=1}^{N} V(t - \Delta t_k)\right] \\ &= \text{sinc}(\omega T_0/2) * \frac{1}{N} \sum_{k=1}^{N} \mathcal{F}[V(t - \Delta t_k)]. \end{aligned}$$

We can rearrange the last line using the time shifting property of the Fourier transform,

$$\mathcal{F}[f(t - t_0)] = e^{-i\omega t_0} \mathcal{F}[f(t)],$$

yielding the general form of a signal that has been averaged N times:

$$\widetilde{V}_{\text{avg}}(\omega) = \left[\frac{1}{N} \sum_{k=1}^{N} e^{-i\omega \Delta t_k} \widetilde{V}(\omega)\right] * \text{sinc}(\omega T_0/2). \tag{2}$$

It remains to examine the sum of complex exponentials in equation (2). Disregarding the constant phase factor, the sum is plotted in Fig. 8. We see it takes the form of a frequency comb, the sharper the greater the number N of averaged traces.

We see from eqn (2) that in the frequency domain, averaging is a two-step procedure. First, the spectrum of the original signal is multiplied by the comb depicted in Fig. 8. If the

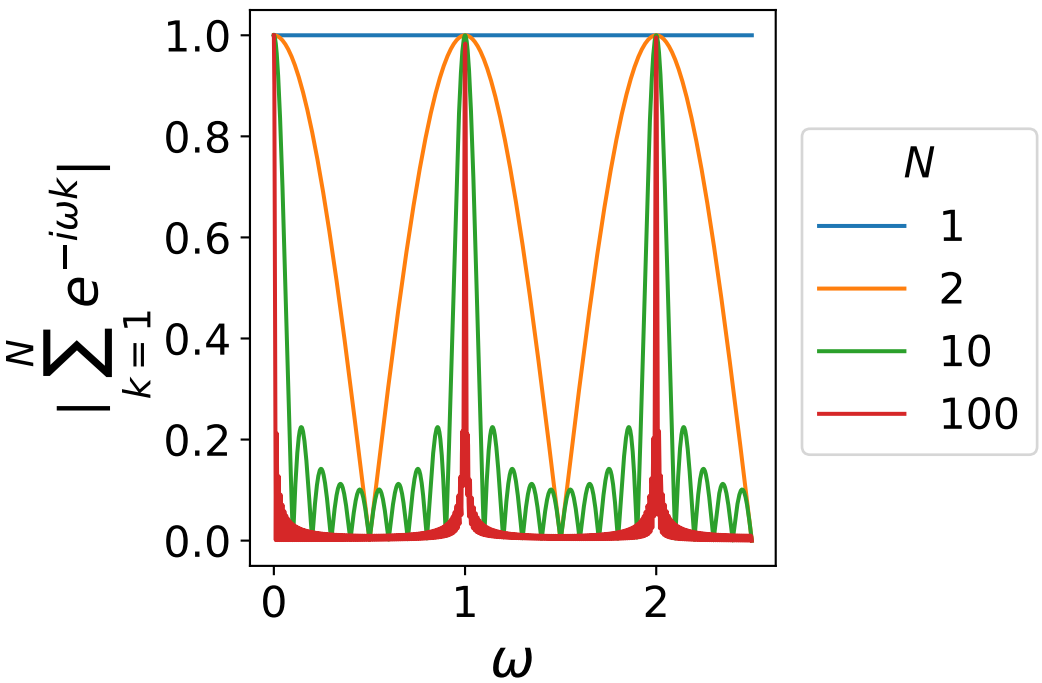

Figure 8: The frequency comb arises when averaging different segments of a measured signal. Before the segments can be averaged together, they have to be time shifted so they overlap; the Fourier transform of time shifting gives the exponentials that sum to form the frequency comb depicted here.

desired signal occurs at the frequency corresponding to one of the comb's peaks (or its harmonics), then the effect of the comb is to bandpass-filter the signal and attenuate the noise away from the desired signal. As stated above, the greater the number N of averages, the sharper the comb, i.e., the lower the bandwidth of the bandpass filter.

The second step to the procedure is the convolution with $\mathrm{sinc}(\omega T_0/2)$. This gives the measured spectrum a linewidth essentially independent of the number of averages. Underneath the sinc, however, the noise will be attenuated through the action of the bandpass filter (or frequency comb) that derives its sharpness from the high number of averages.

Degaussing The hardware used for degaussing in this experiment is as follows. Each of the four shielding layers has its own 12-turn degaussing coil. These are connected to a PCB with a set of relays that can connect each of them to the degaussing current, but only one at a time. During degaussing, the outer layer is always connected first, followed by the second outermost layer, followed by the innermost metglas layer, followed by the mu-metal layer. Since the residual field is primarily determined by the innermost layer, that one is degaussed last lest it be magnetized by accident during the degaussing of the other layers. The relays PCB is in turn connected to an isolation transformer for the purpose of DC blocking. We have considered alternate forms of DC blocking, such as a capacitor, but the required capacitance was too high at the low frequencies typically used in degaussing (down to 1 Hz). The transformer is driven by a power amplifier (Kepco BOP36-12M), which in turn is driven by a home-built DAC board. figure of connections

A wide rage of degaussing waveforms have been used in the literature; for example, linear or polynomial envelopes [REF] or envelopes specifically optimized for a linear decrease in the magnetization of the ferromagnetic material [REF]. In this experiment, I use a finite number of sinusoidal degaussing pulses. The pulses can optionally be separated by and adjustable delay to probe the evolution of the residual field from pulse to pulse. Each subsequent pulse has its amplitude decreased by a constant factor compared with the preceding pulse; thus our protocol is similar to, but not identical to, using an exponentially decreasing envelope multiplying a sine wave. fig: parameters of the degaussing waveform

Despite spending significant amounts of time on optimizing the degaussing parameters,

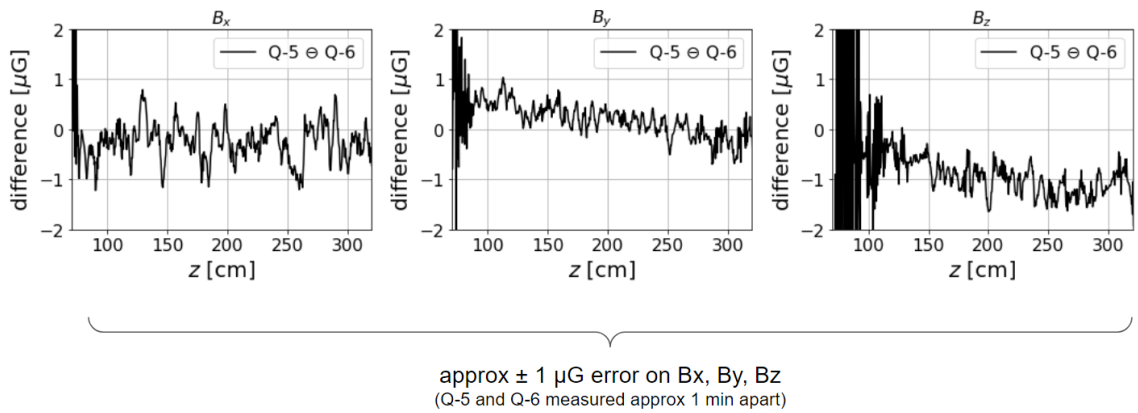

Figure 9: The repeatability of the measurement, shown as the difference between two measurements, one taken immediately after the other.

I am only able to claim the following. The degaussing frequency must be slow enough that the skin effect does not significantly affect the degaussing; the initial amplitude has to be large enough to drive the shield into saturation; the final amplitude has to be small enough that the shield has recovered from saturation and is circling near a point around the origin of the hysteresis curve; the number of degaussing pulses has to be large enough so that the decrease in amplitude is gradual. The exact values of the parameters mentioned in the preceding statements need to be found experimentally for the shield geometry used; fig: degaussing optimization.

Exploration of these parameters is further complicated by the fact that there are four layers to degauss and degaussing one may inadvertently magnetize the others. I have tried degaussing several or all layers in parallel, but have not found any significant improvement and thus have not further explored that method. However, while building the individual metglas layers, I have measured the magnetization of each layer before and after degaussing; fig: data on individual layers before/after degaussing

A further difficulty with studying the degaussing process is the fact that it is not obvious how to put the magnetic shield in a known 'non-degaussed' initial state. In an attempt to optimize the degaussing parameters, I have tried to use a single large-amplitude sinusoidal pulse to magnetize the shield, but it is not clear to what extent this was successful and reproducible.

While degaussing, one may observe the shapes of the hysteresis loops (see example in

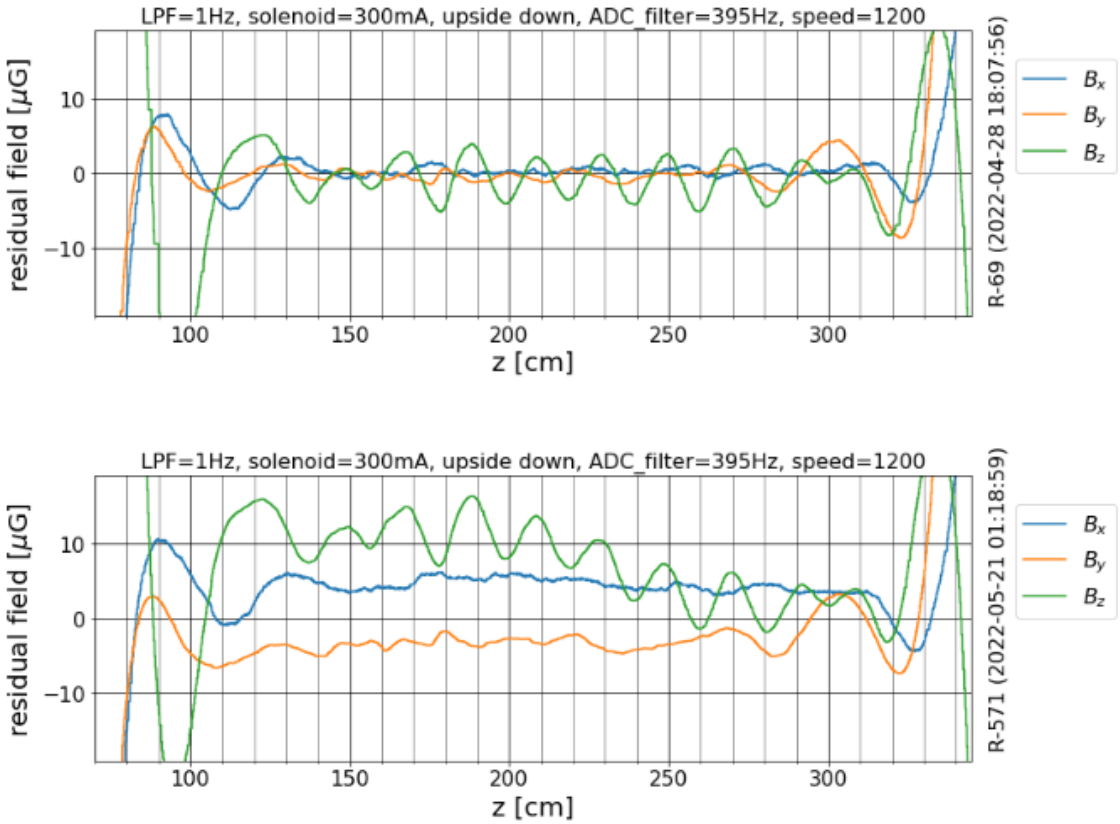

Figure 10: The magnetic field ***a.*** immediately after shimming, and ***b.*** three weeks later.

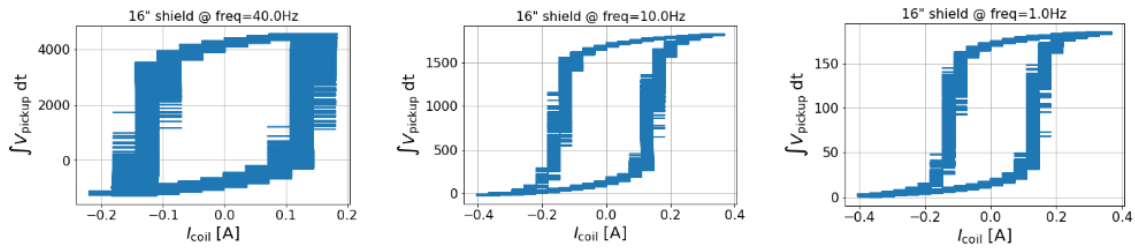

Figure 11: Hysteresis loops observed while degaussing the 16" metglas layer.

Fig. 11) as a function of the degaussing amplitude. These are a useful tool to verify that degaussing is driving the shield into saturation, which we believe essential for the success of the procedure. However, it is difficult to correlate the degree of saturation to its effect on the effectiveness of degaussing. Future work may be done establish the nature of this correlation, which may help optimize the degaussing procedure.

Given the importance of the long-term stability of the magnetic fields inside the shield, it is interesting to explore how stable the degaussing process is, i.e., how much does the field change when degaussing repeatedly, or what happens when the field has drifted away from the degaussed value and the shield is degaussed again. figure

Shim coils Once the shield has been degaussed as well as is possible, the remaining magnetic field has to be cancelled using the system of shim coils. The accuracy of this depends on the measurement of the residual field, as well as on the geometry of the coils themselves. During shield characterization, we have access to the center of the shield and can measure the fields to within 1–2 μG. Thus the limit is set by the number of coil segments and their size.

The shimming process consists of two steps. First, the individual field patterns generated by each of the 80 shim coils have to be measured. I do this by flowing a constant current through the coil under test, measuring the magnetic field, then reversing the direction of the current and repeating the measurement. By subtracting the two measurements, applying smoothing, and masking the parts of the curve where the coil field is expected to be negligible fig: example calibration curve. Once the calibration curves for all the coils are known, I use a simple least-squares minimization routine to determine the optimal currents to use so as to minimize the magnitude of the residual field inside the shield. While I could limit

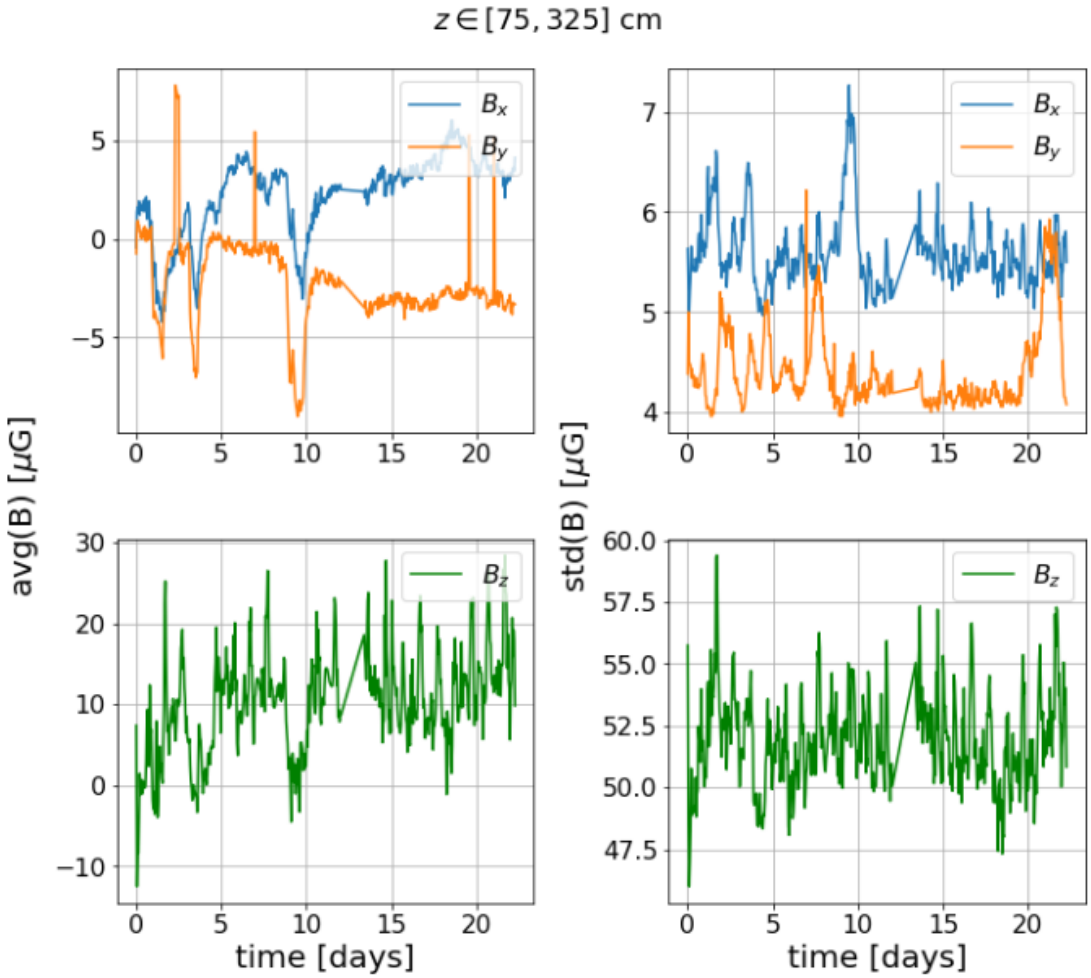

Figure 12: Average and RMS magnetic field in the central 2.5 m of the shield over the course of three weeks.

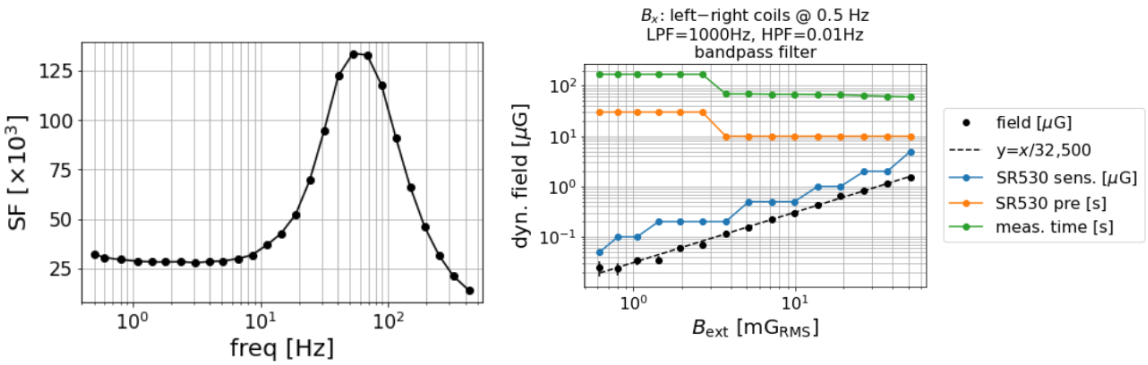

Figure 13: Dynamic shielding factor vs frequency and amplitude of the external field perturbation.

the optimization routine to only minimize the field along the vertical axis, this being the component we are most sensitive to in the molecular beam experiment, I do not anticipate a significant improvement in the shimming by doing so given that the three dimensions of the magnetic field accessed by the coils are orthogonal to each other.

1.5 Performance and limitations

Residual field The design requirements for the shield specified a 10 μG maximum average residual field. This has been met; see Fig. 10 for a typical measurement of the fields inside after shimming.

Of particular interest is how the fields develop after the shimming procedure. To this end, I have followed their evolution over the course of three weeks, finding that neither the average nor the RMS value of any of the two transverse components do not exceed the specified 10 μG (see Fig 12). The data in that plot has been obtained by measuring the fields every hour while keeping all the controllable parameters of the shield constant. Unfortunately, at the time of the measurement, I did not have available the means to monitor the magnetic field external to the magnetic shield, or at any other point inside the shield except on-axis.

Since degaussing is used to significantly reduce the internal magnetic field, it is of interest to observe whether degaussing could be used to periodically restore the initial low-field conditions even after shimming the fields. To estimate the effect of periodic degaussing, I have alternated measuring the field and degaussing; see results in Fig. 14. Unfortunately, it

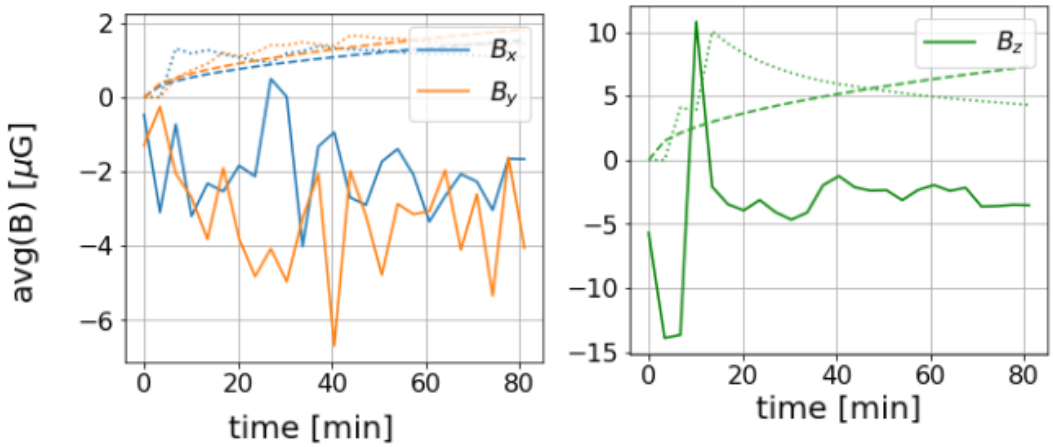

Figure 14: Average internal magnetic field while degaussing every 200 seconds.

appears that degaussing leaves the shield in a state of a small but significant internal field. Since the shield is stable enough, I believe better performance is obtained by degaussing only at the beginning of measurements, then shimming the shield, and letting it stay that way.

It is true that just because the shield stayed in a low-field condition for three weeks once, it is not certain it will always stay that way for as long. In other words, it would be useful to find a model for predicting how long the shield stays this way, or what the probability is that is has exceeded 10 μG. If we model the average internal magnetic field as a random walk, then we can associate a diffusion constant to this process and calculate the probabilities. Include the calculations for random walk. Of course, there is not physical reason why the growth in internal magnetic field should be a random walk; it would be of interest to compare other possible stochastic models for this behaviour, e.g. using Allan deviation.

It would also be interesting to measure the noise spectrum inside the shield to see if it is of a similar nature (i.e., maybe random walk) as the slow drifts in the residual field.

Dynamic shielding In addition to the static internal field, it is also important to ensure the exteral magnetic field noise is sufficiently attenuated inside. My measurements of the dynamic shielding factor place it around 30,000 at DC (see Fig. 13), which should reduce the

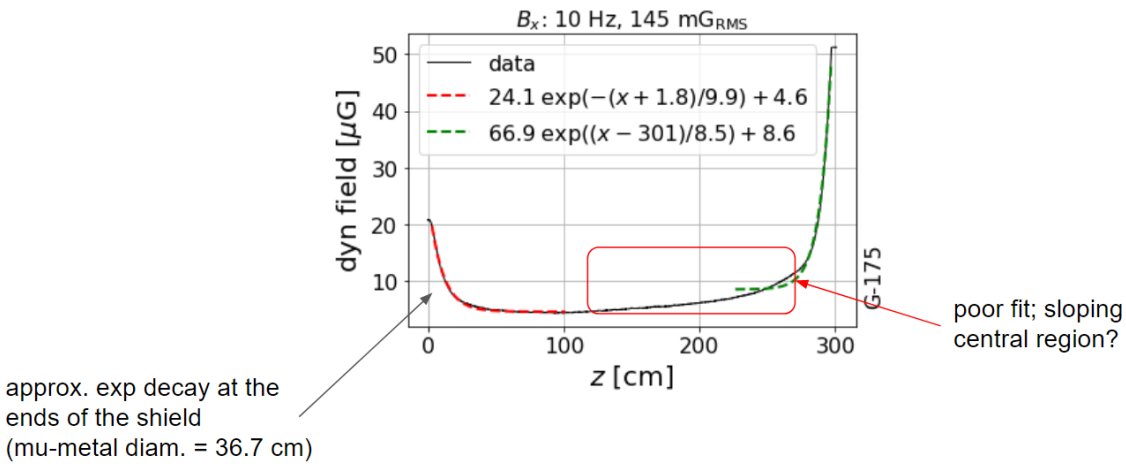

Figure 15: Spatial dependence of dynamic SF.

expected 1 mG variations in the lab field well below the level at which they would impact the molecular-beam measurements. The frequency dependence of the dynamic SF shown in that figure shows an increase in shielding as frequency increases, which makes sense in light of skin effect considerations. For mumetal of 1 mm thickness and relative permeability of around 50,000, a frequency of XX Hz would correspond to a skin depth comparable to the inner-layer thickness. (The decrease in shielding once the maximum is achieved is likely not physical and reflects some limitation of my measuring setup. I have, however, been so far unable to find out what causes this.)

The spatial dependence of the dynamic SF is as shown in Fig. 15. It will be noted that the pattern is asymmetric, as if one side of the shield works better than the other. That is also unlikely to be the case. Further work may establish what causes this observation. One possibility is that in addition to attenuating the external excitation, the shield also changes the direction of the external field. By measuring only one component of the shielding, we are unable to distinguish between attenuation and change of direction; a three-dimensional measurement may clarify the situation. Another possibility is that the asymmetry is due to some relaxation (or low-pass filtering) in the measurement, such that reversing the direction of sensor motion may change the results. It could also be the case that despite my careful axis centering, the transverse position of the sensor changes along the length of the shield, such that the sensor is closer to the walls of the shield at one end. The fields are likely

larger near the walls of the shield, but more work is needed to determine the transverse profile of the dynamic shielding factor.

The dynamic SF measurements on the final shield were all done with the external excitation caused by the left-right pair of rectangular coils, corresponding to an approximately horizontal transverse external field. We expect from the cylindrical geometry of the shield that the longitudinal shielding is significantly worse, though we have not yet measured it. Similarly, it would be important to measure the shielding in the vertical transverse direction.

The dynamic shielding is likely affected by the magnitude of the external stationary fields. In large lab field, the outer layers of the shield will saturate and become less effective. On the other hand, given the relatively small size of the fields at the inner layers of the shield, the shim coils are unlikely to affect the dynamic shielding. However, neither of these claims have as yet been explicitly verified with the final version of the magnetic shield.

1.6 Discussion

While the dynamic shielding follows the approximate theoretical model, the static shielding does not. It is not clear why degaussing achieves such good shielding, nor is it known what limits it from working even better. It appears clear from a simple model that degaussing magnetizes the shield in such a way as to oppose the external magnetic fields, that being the configuration of lowest total energy.